

Programme	B. Sc. Mathematics Honours			
Course Code	MAT6CJ306 / MAT8MN306			
Course Title	METHODS OF DIFFERENTIAL EQUATIONS			
Type of Course	Major			
Semester	VI			
Academic Level	300-399			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Foundations of basic calculus (0-99 level)			
Course Summary	The course enhances the skill to solve ordinary differential equation using specific methods analytically and computationally for first and higher order differential equations.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Classify and solve first order differential equation by applying appropriate methods	Ap	C	Internal Exam/ Assignment/Seminar/ Viva/End Sem Exam
CO2	Apply different methods to solve higher order homogeneous and non-homogeneous linear differential equations with constant coefficients	Ap	C	Internal Exam/ Assignment/Seminar/ Viva/End Sem Exam
CO3	Use Laplace transform and inverse Laplace transform to solve linear differential equations	Ap	C	Internal Exam/ Assignment/Seminar/ Viva/End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook	Dennis G. Zill , A First Course in Differential Equations with Modeling Applications 10 th Edn, Cengage Learning (2012) ISBN-13 978-1111827052			
Module	Un it	Content	Hrs (60)	Marks Ext: 70
I	First order differential equations		14	Min.15
		Quick review of Introduction to differential equations (Definitions only)		
	1	2.1.1-Direction Fields		
	2	2.1.2 - Autonomous First-Order DEs		
	3	2.2 - Separable Equations		
	4	2.3 - Linear Equations		
	5	2.4- Exact Equations		
	6	2.5- Solutions by Substitutions		
	7	Problems from the above sections		
II	Higher-Order Differential Equations		12	Min.15
	8	4.1.1 Initial-Value and Boundary-Value Problems		
	9	4.1.2 Homogeneous Equations (proof of Theorems 4.1.2 and 4.1.5 are optional)		
	10	4.1.3 Nonhomogeneous Equations		
	11	4.2 Reduction of Order		
	12	4.3 Homogeneous Linear Equations with Constant Coefficients		
III	Higher-Order Differential Equations (Cont..)		14	Min.20
	13	4.4 -Undetermined Coefficients—Superposition Approach (up to and including Example 9)		
	14	4.5 - Undetermined Coefficients—Annihilator Approach (up to and including Example 3)		
	15	4.5 - Undetermined Coefficients—Annihilator Approach (all the topics after Example 3)		
	16	4.6- Variation of Parameters		
	17	4.7 - Cauchy-Euler Equation (up to and including Example 4)		
	18	4.7 - Cauchy-Euler Equation (all the topics after Example 4)		
	19	4.9 - Solving Systems of Linear DEs by Elimination		
IV	Laplace Transforms		8	Min.10
	20	7.1 Definition of the Laplace Transforms (proof of Theorems 7.1.2 and 7.1.3 are optional)		
	21	7.2.1 Inverse Transforms		
	22	7.2.2 Transforms of Derivatives		
V	Open Ended: Mastering differential equation using software		12	
	IVP and BVP Problem-solving using mathematical software like Sage/Python/ Mathematica/Matlab/ Maple/Scilab etc (Instructor may choose any software appropriately) Suggestions: ● Plotting solution curves -2 hrs			

	• Solve first order initial value problems -2 hrs • Solve second order initial value problems -2 hrs • Plot Laplace transform of given function -2 hrs • find Laplace transform and inverse Laplace transform - 2 hrs • Solve the initial value problem using Laplace transform -2 hrs		
	References 1. G. F. Simmons and S. G. Krantz, Differential Equations: Theory, Technique, and Practice, McGraw Hill (2006), ISBN-13. 978-0072863154 2. E. A. Coddington, An Introduction to Ordinary Differential Equations, Prentice Hall India (2009). ISBN: 9788120303614 3. E. Boyce , Richard C. DiPrima, Douglas B Meade, Elementary Differential Equations and Boundary Value Problems, 11 Edn. William John Wiley & Sons (2017) ISBN: 1119169879 4. William F. Trench, <u>Elementary Differential Equations with Boundary Value Problems</u>, S.Chand (G/L) & Company Ltd (2013) ISBN 13: 9780534368418. 5. S. L. Ross, Differential Equations, 3rd edition, Wiley India, (2007) ISBN-13. 978-8126515370 6. Martha L. Abell, James P. Braselton, Differential Equations with Mathematica, 5th edn. Elsevier Science Publishing Co Inc (2022), ISBN: 9780128241608 7. Amit Saha, Doing Math with Python", No Starch Press, US . (2015), ISBN 13 978-1593276409		

***Optional topics are exempted for end semester examination.**

****70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module.**

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	2	1	3	0	3	0	3	0	0
CO 2	2	3	1	2	3	0	3	0	3	0	0
CO 3	2	1	3	3	3	0	3	0	3	0	0

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓